

GHOST INFRASTRUCTURE: Development Log

This is a working log for the Ghost Infrastructure project, tracking whether 19th and 20th century coal mine and worker colony locations in Bochum still predict present day 15 minute city walking accessibility in the Ruhr Valley. Entries were written as the work happened, not pieced together afterward. Design choices, dead ends, and fixes all got recorded in the moment, including the reversed result that overturned the original hypothesis.

The log follows the project's actual build order: formulating the research question, digitizing historical mine and colony locations, building the network based walking accessibility model, running the statistical tests, investigating the reversed result, producing the final figures, and later adding robustness checks and a second city replication. Every number and p value below is what the corresponding script produced when run.

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Entry 1

Project name: Ghost Infrastructure. The goal is to test whether 19th and 20th century coal mines and worker colonies (Zechensiedlungen) in the Ruhr Valley still determine present day walking accessibility, more than 50 years after the industry's decline and the 1974 closure of the last mine. The original scope included steel factories and historical rail lines, but the data actually collected covers only coal shafts and worker housing, since these were the categories with sufficiently complete archival records to digitize reliably. Steel factories and rail lines went into the research paper's future work list instead of being added on an incomplete basis.

Most existing literature on Ruhr Valley industrial heritage is descriptive, not quantitative: historical narrative, heritage preservation, or planning discussion. Separately, 15 minute city research typically explains accessibility gaps using present day variables (income, demographics, age) without accounting for historical industrial geography. This project combines digitized historical industrial locations with a network based

accessibility model and spatial statistics to test whether present day accessibility gaps trace back to historical industrial geography.

Research question: does the historical geography of the Ruhr Valley coal and steel industry (mine locations, worker colonies, historical rail lines) still predict which present day neighborhoods fall outside a 15 minute walking radius of essential services?

4 tasks follow from this: (1) digitize and georeference historical industrial era maps to locate coal shafts and worker housing (historical transport lines were deferred to future work); (2) build a present day walking network model generating network distance accessibility to daily services, clinics, food shops, primary schools, and parks, using current road data; (3) apply spatial statistics (Local Moran's I, hot spot analysis) to test whether low accessibility areas cluster near historical industrial sites or just appear at random; (4) produce a map overlay of historical industrial geography against present day accessibility to test the "ghost infrastructure" hypothesis directly.

Bochum, North Rhine-Westphalia, was selected as the study area: a farming town before its mid 19th century industrialization into a major coal and steel city through the 1950s. The official Industrial Heritage Route (route-industriekultur.ruhr) provided an initial list of known worker housing colonies within city limits. An initial search bounding box (51.42°N–51.53°N, 7.13°E–7.30°E) was set, to be refined once actual historical map data was available.

Entry 2

Built the historical site data layer first, since all subsequent work depends on it. Coal mine coordinates were collected from Mindat.org (manually, page by page, since no bulk export is available); worker colony coordinates were compiled from German heritage archive pages, Wikipedia, the Route Industriekultur portal, and ruhr-bauten.de.

The result is 13 coal mines and 4 worker colonies, stored as 2 separate GIS layers reflecting their distinct categories. 1 steelworker neighborhood, Stahlhausen (built by the Bochumer Verein steel company), was excluded from the worker colony file since it belongs to steel production, not coal. Both layers were converted to GeoPackage format (EPSG:4326) via GeoPandas for use in QGIS and subsequent network analysis.

Entry 3

Downloaded Bochum's current pedestrian street network via OSMnx: 69,393 nodes and 169,668 edges. Retrieved 786 essential service locations from the same source, covering hospitals, clinics, pharmacies, primary schools, kindergartens, supermarkets, corner stores, and parks.

Built the accessibility model using true network distance (Dijkstra's algorithm) instead of straight line buffers. Straight line distance overestimates walkability, because it ignores building and block layout. Each node's accessibility is computed as the true walking network distance to the nearest of the 786 service locations.

Entry 4

Ran the core statistical test with the completed historical site data (17 locations) and the full accessibility model (69,393 nodes): a Welch's t test comparing distance to the nearest historical industrial site between low accessibility and high accessibility nodes.

The result reversed the original hypothesis. The test found a highly significant difference ($t=42.887$, $p<0.00001$) in the opposite direction from what was expected: low accessibility nodes are farther from historical industrial sites (mean 1,984 m) than high accessibility nodes (mean 1,450 m). Proximity to historical coal mining sites is therefore associated with better present day accessibility, not worse.

This gets written up as the main finding, not thrown out as an error. Historical industrial development had to be built around dense worker housing, and that dense, central urban layout still seems to hold onto better service density and street connectivity today than the more outer areas do. Call it a "path dependency of centrality." It is not a "path dependency of neglect."

Entry 5

Before accepting the reversed result, tested an obvious confound: historical mines may simply sit close to Bochum's city center, which independently has strong walking accessibility, so the apparent historical site effect might just be standing in for city center proximity, not a real effect on its own.

The correlation between distance to historical site and distance to city center is low ($r=0.063$), indicating the 2 are largely independent. A logistic regression predicting accessibility from both distances jointly retained a statistically significant historical site effect (coefficient=-0.0005, $p<0.001$) after controlling for city center distance. So the historical site effect is a real, separate signal, not just a restatement of the known city center advantage.

Entry 6

Returned to a planned but not yet completed task: the project's original scope required a Local Moran's I / hot spot analysis to test whether low accessibility areas cluster spatially, since the Entry 4–5 tests only address whether distance differs between groups on average, not whether that difference is spatially clustered.

Built a row standardized k nearest neighbor ($k=8$) weights matrix across all 69,393 nodes and computed Local Moran's I on the binary accessibility flag using 99 permutations (`libpysal`, `esda`; seed=42; significance at $p<0.05$).

Results: mean local I = 0.923; 10,266 of 69,393 nodes (14.8%) showed statistically significant spatial clustering. Of these, 9,568 were Low Low (cold spot) and 698 were High Low (spatial outlier), with no significant High High or Low High clusters. Nodes in significant Low Low clusters averaged 1,992.3 m from the nearest historical site, versus 1,447.1 m for non significant nodes; 97.1% of all low accessibility nodes fell within a significant Low Low cluster.

A second, independent method backs up the Entry 4 finding: low accessibility is not randomly scattered. It forms real, connected clusters that sit measurably farther from historical industrial sites. Saved the cluster map and `spatial_clustering_lisa.py` to the repository and recorded the results in Section 4.4 of `GI_Research_Paper.md`.

Entry 7

The map overlay visualization (all 69,393 nodes by accessibility, plotted against historical mine and colony points) initially rendered as a blank canvas with a single point in the corner. Diagnosed as a coordinate reference system mismatch: the street network had been saved in EPSG:32632 (UTM Zone 32N, needed for earlier metric distance calculations), while the historical site layers remained in EPSG:4326. Plotting both

without reprojecting to a common CRS caused the geographic degree coordinates to collapse to a single pixel next to the much larger UTM coordinate values. Reprojected all layers to EPSG:4326 before rendering, which resolved the issue.

2 further apparent anomalies were found on inspection: only 12 of 13 coal mine markers were visible, and 2 of 4 worker colony markers appeared to overlap. Instead of assuming a code defect, the underlying coordinate data was checked directly. The Mansfeld and Heinrich Gustav mines (Langendreer and Werne) are 1.75 km apart, close enough to merge visually at full city zoom, and the Kolonie Hannover and Am Rübenkamp worker colonies are 0.33 km apart, having been built adjacent to each other to serve the same Hannover mine shafts during overlapping construction phases (1874–1892). Verified via direct Haversine distance calculation between the coordinate pairs. Given Bochum's roughly 14 km extent, both cases are real, genuine historical spatial proximity. Neither is a data or rendering error, so no code changes were needed.

The general practice this set: never just assume an unexpected result (Entry 4) is right, or a bug. Check it against a second, independent method instead (Entry 5). And never assume a strange looking map is a rendering error either. Check it against the raw coordinates first, and only call it real geography once that check backs it up.

Entry 8

Completed the core analysis (statistical tests, confound checks, cluster maps, and figures) and assembled the multi page Streamlit dashboard ([app.py](#)), with separate pages for the study design, historical maps, accessibility results, the main finding, trend analysis, interactive maps, and methodology. Finalized the research paper, project report, and [README.md](#), and pushed the repository to GitHub.

Entry 9

Returned to this project to extend its scope and address documented limitations, selecting it first among the portfolio projects since it requires no Google Earth Engine access (only OSMnx, OpenStreetMap, and GADM boundary data) and already had a defined Future Work list.

First extension: a walking threshold sensitivity check, addressing an item noted as incomplete in the original paper. Wrote [threshold_sensitivity.py](#) to rerun the full accessibility classification, distance calculation, Welch's t test, and logistic regression confound check at 10 minute (750 m) and 20 minute (1,500 m) thresholds, reusing the existing Bochum street network and distance data without new downloads.

The reversed relationship held at both thresholds, and strengthened somewhat at the wider threshold: 10 minute $t=47.062$, $p<0.00001$, $d=0.413$ (versus the original 15 minute $t=42.887$, $d=0.589$); 20 minute $t=32.150$, $d=0.661$. The odds ratio per 100 m closer to a historical site remained stable across all 3 thresholds (4.24%, 4.88%, 4.49%), indicating the original 15 minute result is not an artifact of that specific threshold choice. Recorded in Section 4.5 of [GI_Research_Paper.md](#).

Entry 10

Selected Essen for a second city replication, per the project's own Future Work list, since it lies 15 km northeast of Bochum, shares the same 19th century coal mining history, and has comparably strong heritage documentation.

Initial data collection was blocked by outages in OpenStreetMap's Overpass API and Nominatim services, Wikidata's live endpoints, and a 403 error from Mindat.org (the source used for the Bochum mine list). KuLaDig, North Rhine-Westphalia's official state heritage database, remained available and provided WGS84 coordinates for the major heritage listed mine shafts; German Wikipedia articles for worker colonies also reliably included geo tagged coordinates, unlike the corresponding mine shaft articles.

Digitized 4 coal mines: Zeche Zollverein Schacht XII, Katernberg (shaft operated 1932–1986; part of the wider Zollverein colliery complex founded 1847/51); Zeche Carl Funke, Heisingen (1804–1973); Zeche Vereinigte Helene & Amalie, Altendorf (1843/44–1968); Zeche Pörtingsiepen, Fischlaken (1779–1972). And 4 worker colonies: Siedlung Carl Funke (1900), Mathias-Stinnes-Siedlung, Karnap (1890), Kolonie Zollverein III, Katernberg (1880), and Kolonie Beisen (1902). All 8 points were checked against the GADM v4.1 Germany boundary layer to confirm they fall within Essen's administrative boundary.

This 8 site Essen dataset is considerably smaller than Bochum's 17 sites; Essen's official historical portal lists over 1,700 mining related locations citywide, so this small dataset is written up as a clear limitation. It is not a full picture. The dataset initially used 3 mines and 3 colonies (6 sites); when early statistical checks (Entry 11) showed unstable results attributable to the small sample, 2 further sites were added to reach the current 8 site version.

Downloaded Essen's pedestrian network via a new script, [download_network_essen.py](#) (mirroring the Bochum download logic against the Overpass API): 72,027 nodes, 188,198 edges, and 1,410 raw service facility points, filtered by the same point geometry criteria used for Bochum down to 366 usable service locations.

Entry 11

Ran the Essen replication using a new pipeline script, [run_essen_pipeline.py](#), replicating the Bochum methodology (Entries 3–6) exactly: 1,125 m network distance threshold, Welch's t test, Essen Hauptbahnhof as the city center reference, and k=8 Local Moran's I at 99 permutations with seed 42.

88.5% of Essen's 72,027 nodes fall within a 15 minute walk of at least 1 service (versus 85.8% in Bochum). The reversed relationship replicated: low accessibility nodes (n=8,267) average 3,693 m from the nearest historical site versus 3,130 m for high accessibility nodes (n=63,760); Welch's $t=24.731$, $p<0.00001$, Cohen's $d=0.338$ (smaller than Bochum's 0.589, but the same direction and highly significant). Local Moran's I results were also consistent: mean local I=0.917 (versus 0.923 in Bochum), with 95.5% of low accessibility nodes in significant Low Low clusters (versus 97.1% in Bochum) and no significant High High clusters in either city.

The confound check did not replicate. The correlation between historical site distance and city center distance is $r=0.405$ in Essen, substantially higher than Bochum's $r=0.063$. In the joint logistic regression, the historical site coefficient's sign reversed relative to Bochum: $+0.0001$ ($p<0.00001$) in Essen versus -0.0005 in Bochum, meaning that once city center distance is controlled for, greater distance from a historical site is associated with higher, not lower, odds of accessibility in Essen.

Checked this against dataset size directly: the original 6 site Essen dataset gave $r=0.475$ for the same correlation; expanding to 8 sites brought it down to $r=0.405$, but the sign flip stayed. So sample size seems to partly drive the size of the correlation, but it does not fully explain why the sign flipped.

Entry 12

Followed the same approach as Entry 7 when this unexpected result showed up: dug into it, without dismissing it or forcing a preferred explanation on it. 2 candidate explanations were tested directly, side by side, instead of just picking whichever one fit the story better.

First possibility: Essen's historical mining sites may simply be geographically closer to its city center than Bochum's more dispersed sites. Second possibility: the smaller Essen historical site sample (8 versus Bochum's 17) makes this specific correlation more sensitive to sampling variability.

The 6 to 8 site comparison ($r=0.475$ to $r=0.405$) shows sample size affects the correlation's magnitude, but does not by itself account for the full sign reversal, so both explanations went into Sections 4.6 and 7 of [GI_Research_Paper.md](#), side by side, instead of picking just 1 cause. Expanding the Essen dataset further is listed as future work, since locating precise coordinates for the remaining, less documented historical sites was not feasible within the current scope.

Overall conclusion from the 2 city comparison: the core finding, that low accessibility clusters spatially and correlates with distance from historical industrial sites, replicates in both cities. The additional claim that this effect is fully independent of city center proximity holds in Bochum but not in Essen, so for now, that piece is written up as specific to Bochum, until more data comes in to say otherwise.

Entry 13

Identified an inconsistency between the 2 cities' interactive map implementations: the Bochum overlay map still used an older QGIS2Web export from early in the project, while the Essen map had been built directly in Python with Folium. Rebuilt [outputs/maps/bochum_interactive_map.html](#) from the underlying GeoPackage files (accessibility metrics, coal mine and colony coordinates, GADM boundary), matching the Essen map's style: dark CartoDB base tiles, orange markers for mines, light blue markers for worker colonies, with popups showing site name, district, and operational dates.

Bochum has 9,858 low accessibility nodes; rendering these as individual markers would be impractical in the browser, so they are rendered as a heatmap density layer instead (Essen's 8,267 equivalent nodes use the same approach). Both cities' administrative boundaries use the same teal and cream color scheme.

1 difference remains between the 2 cities' popups: Essen's include a source URL (KuLaDig/Wikipedia, collected during digitization), while Bochum's original data did not retain a citation field, so Bochum popups show name, district, and operational years only. No coordinate or date values were changed during this rebuild. The underlying data (digitized coal mine and colony coordinates, the 69,393 node network, and the boundary shape) is unchanged; this was a rendering pipeline change only, replacing the QGIS2Web export with a single HTML file generated directly from the project's own GeoPackages. Updated [dashboard/pages/6_Interactive_Maps.py](#) to embed the new map via [components.html](#), matching the Essen page, and removed references to QGIS2Web from [README.md](#).

Entry 14

Revisited the Local Moran's I analysis from Entry 6, which used 99 permutations. That was adequate at the time but low for a more stable p value estimate at this sample size (69,393 nodes). Increased [PERMUTATIONS](#) in [spatial_clustering_lisa.py](#) from 99 to 999 and reran the analysis.

Results were essentially unchanged: 10,273 significant nodes (versus 10,266 previously), with 9,571 Low Low and 702 High Low (versus 9,568 and 698). Mean local I remained 0.923, and the headline 97.1% figure was

unchanged. So the original 99 permutation result was already stable, no fix needed there.

Also tested an alternative neighbor definition. Instead of KNN (k=8) by straight line distance, `spatial_clustering_lisa_network.py` defines neighbors as nodes reachable within 750 m along the actual street network. This produced a very different result: 96.9% of nodes significant (versus 14.8% under KNN), and this time High High hot spots dominated (53,421 of 67,210 significant nodes), not Low Low cold spots. The likely cause: a fixed 750 m network buffer gives each node a wildly different neighbor count (potentially hundreds in dense areas), while KNN keeps that count fixed at 8. Since this is not a like for like comparison to the original KNN based result, it is not used as a validated finding and is recorded as an unfinished side experiment; a fair comparison would require calibrating the network distance cutoff to yield approximately 8 neighbors per node on average.

Addressed the non independence of adjacent street nodes directly via `network_block_bootstrap.py`: partitioned the network into 2,598 contiguous blocks (grown outward from random starting nodes toward a target of 500 connected nodes each) and resampled whole blocks with replacement 999 times, refitting the difference of means and logistic regression each iteration. The observed distance difference (534.25 m) produced a 95% bootstrap confidence interval of [199.90, 885.16] m, excluding 0; both logistic regression coefficients also excluded 0 in their respective intervals. So the resampling unit here is the network's own connected structure, not each of the 69,393 nodes taken as independent, and the result held up. Added this to Section 7 (Limitations) of `GI_Research_Paper.md`, alongside the existing note on node non independence, and listed the network distance LISA attempt and a block size sensitivity check under Section 8 (Future Work) as unfinished items.

Entry 15

Completed the block size sensitivity check previously listed as future work. Wrote `network_block_bootstrap_sensitivity.py`, rerunning the Entry 14 network block bootstrap at target block sizes of 250 and 1,000 nodes, alongside the original 500, to test whether the reported confidence interval was specific to the 500 node choice.

Results: 250 nodes, CI [292.21, 803.04]; 500 nodes, CI [199.90, 885.16] (identical to the Entry 14 result, as expected since this is the same 500 node target under the same seed); 1,000 nodes, CI [135.78, 884.20]. All 3 exclude 0 and retain the same sign on the historical site coefficient; the interval widens somewhat as block size increases, consistent with fewer, larger resampling units. Moved this from Future Work into Section 7 as a completed robustness result. (Correction: this entry originally reported different, non-reproducible numbers here. The cause was `make_network_blocks()` drawing its random starting node from an unsorted Python set; that set's iteration order depends on the process's string hash seed, which varies run to run regardless of the fixed numpy seed, so the block partition itself was not actually reproducible. Fixed by sorting the node IDs before the random draw in both `network_block_bootstrap.py` and `network_block_bootstrap_sensitivity.py`; the numbers above are from the corrected, reproducible run.)

Also corrected a wording error in the Limitations section, which described bootstrap blocks as averaging roughly 500 nodes each; the script's own output shows the true average is approximately 27 nodes per block, since the pedestrian network's many small disconnected fragments prevent most blocks from growing near the 500 node target. Added a clarifying note in Section 3.3 that the accessibility flag is a single binary indicator, at least 1 service of any kind within 15 minutes. It is not a per category or weighted score. The original text had not made this clear enough.

Entry 16

While reviewing the Essen historical site count (quoted throughout the paper, this log, and `essen_results.json` as "8 sites: 4 mines + 4 colonies"), checked the current `essen_coal_mines.gpkg` and `essen_zechensiedlungen.gpkg` files directly and found only 6 sites (3 mines, 3 colonies). Zeche Pörtingsiepen and Kolonie Beisen, both digitized and named in Section 3.7 of the paper, were missing from the current GeoPackage files, most likely because an earlier export was saved over the final one without being noticed. `essen_accessibility_with_distance.gpkg` and `essen_results.json` still contained the correct 8 site derived numbers from when the pipeline was last run against the complete data, so the paper's reported figures were never incorrect. But rerunning `run_essen_pipeline.py` against the repository as it stood at that moment would have brought back the earlier 6 site result ($r=0.475$), not the reported 8 site result ($r=0.405$).

Reverified both missing sites' locations: Zeche Pörtingsiepen (Fischlaken, 51.394868°N, 7.043953°E; German Wikipedia and historischesportal.essen.de) and Kolonie Beisen (Katernberg, 51.49247°N, 7.06959°E; KuLaDig, associated with Zeche Zollverein). Added both back into the 2 GeoPackage files with the same schema as the existing entries, then reran the historical distance, correlation, logistic regression, and Local Moran's I steps of `run_essen_pipeline.py` against the corrected 8 site data. Result: `mean_dist_low`=3,693.4 m, `mean_dist_high`=3,129.9 m, `t`=24.732, `d`=0.338, `r`=0.4048, logistic coefficient=+0.000114, mean local I=0.917, 95.5% of low accessibility nodes in significant Low Low clusters. The numbers match the paper and `essen_results.json` almost exactly, so these are the correct 2 sites, not a rough approximation. Regenerated `essen_accessibility_with_distance.gpkg` and `essen_results.json` from the corrected files. The full pipeline can now be reproduced directly from the repository's current data. It no longer depends on a frozen, non reproducible output.